YOUR TECHNOLOGY & INNOVATION PARTNER

Advanced AI Training - Session 4

Advanced Computer Vision

Training Objective

- Explore the Evolution of Computer Vision
- Recap Key Concepts of Convolutional Neural Networks (CNNs)
- A Quick Guide to Understanding Vision Transformers
- Showcase of Computer Vision Projects at TMA
- Introduction to the Teledeaf Solution by the Solution Center

Overview

- I Evolution of Computer Vision
- II Applications
- III SC Sharing - Teledeaf

Evolution of Computer Vision

Brief History of Computer Vision (CV)

Early 80s

Early Age of CV

1966 - Summer Vision Project

1979 - Neocognitron

1989 - 2014

Foundational CNN

1989 - LeNet (Handwritten Digit)

1998 - LeNet (Character Recognition)

2011 - AlexNet (Object Classification)

2014 - 2016

Deeper Network

2014 - VGGNet

2015 - ResNet, Inception

2016 - 2020

Efficiency Focused Model

2016 - MobileNet

2018 - EfficientNet

2020 & Beyond

Transformer-based Model

2020 - ViT

2021 - CLIP

2022 - BLIP

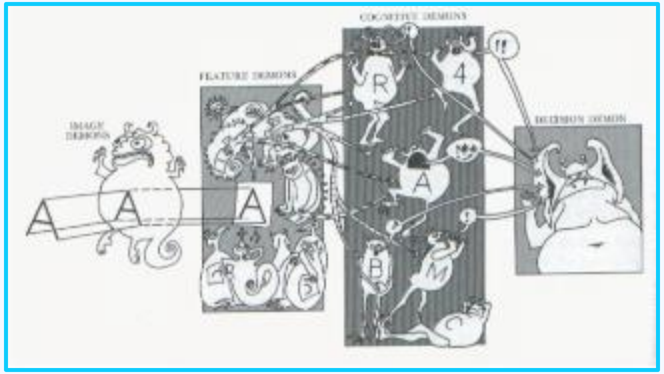
2023 - Multi-modal LLM, Stable Diffusion

Image Processing → CNN → Efficiency → Transformer

Early Age of CV

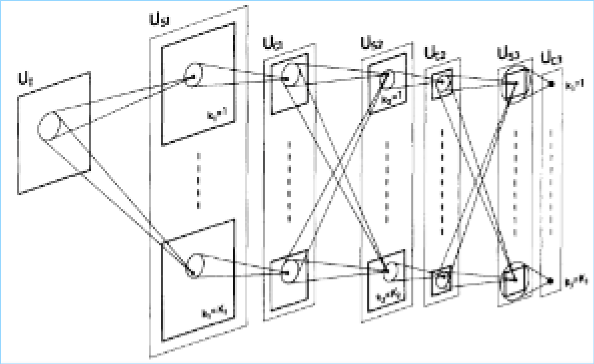
Summer Vision Project

The **Summer Vision Project** (MIT, 1966) aimed to separate foreground from background in images. Though it failed, it revealed the complexity of visual perception and is widely seen as the **starting point of computer vision**.



Neocognitron

The **Neocognitron**, proposed by Fukushima in 1980, laid the foundation for CNNs by recognizing patterns through layered shape detection. Its self-organizing structure enabled position-invariant visual recognition, pioneering automated image understanding.



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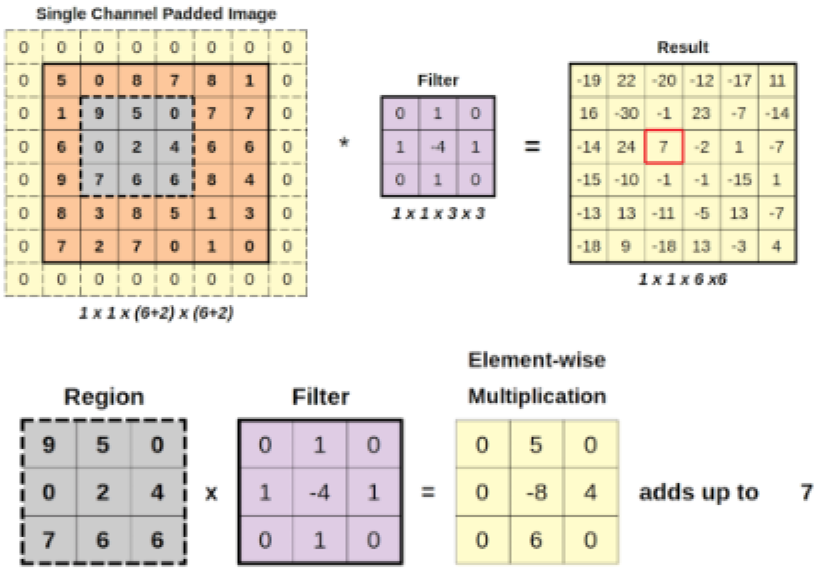
CNN Recap (1/2)

A **Convolutional Neural Network (CNN)** is a deep learning model designed primarily for processing visual data. Its core component, the **convolutional layer**, applies filters (kernels) that **convolve** over the input image to extract local features such as edges, textures, and shapes. These features are passed through successive layers to learn increasingly complex patterns, making CNNs highly effective for tasks like image classification, object detection, and segmentation.



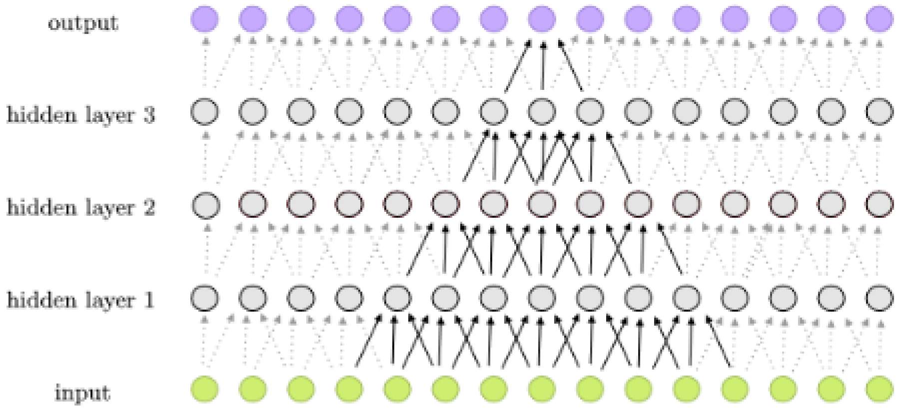
Local Feature Extraction

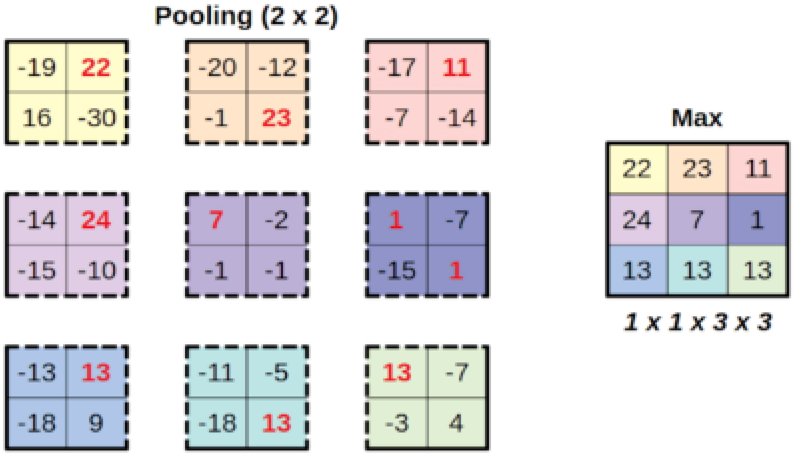
Hierarchical Representation

Translation Invariance

CNN Recap (2/2)

- **Pooling layers** reduce the spatial dimensions of feature maps, typically using operations like max pooling, helping to retain important features while reducing computation and overfitting.
- **Fully connected layers** appear near the end of a CNN and act as a classifier, taking the flattened features and producing the final prediction output.





Foundational CNN (1/3)

Paper - Handwritten Digit Recognition with a Back-Propagation Network

The **Handwritten Digit Recognition with a Back-Propagation Network**, developed in the late 1980s by **Yann LeCun**, was the first successful application of a CNN trained with back-propagation. Inspired by the Neocognitron, it used a **3-layer architecture** to recognize digits **0–9**, laying the groundwork for modern deep learning in image classification.

Classification of handwritten digits

Simple CNN with only 3 layers

Tested on MNIST dataset

Source: Link

Foundational CNN (2/3)

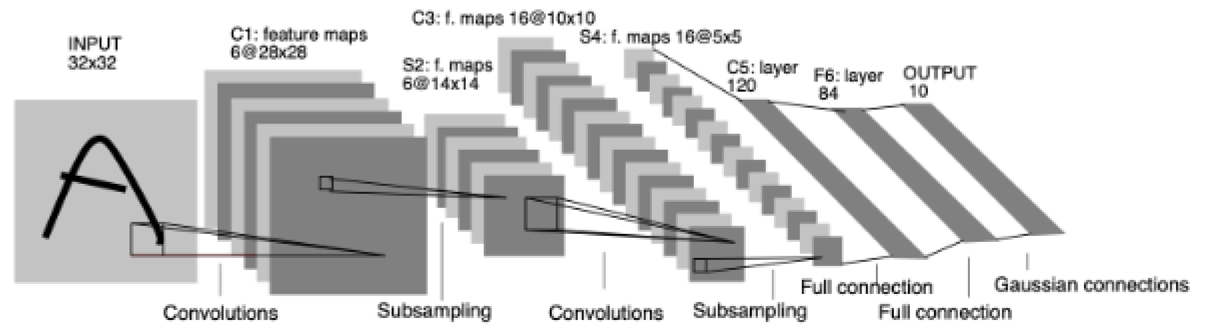
Paper - Gradient Based Learning Applied to Document Recognition

Gradient-Based Learning Applied to Document Recognition, introduced by **Yann LeCun** in 1998, presented **LeNet-5**, a 5-layer CNN designed to recognize characters in documents. Building on earlier digit recognition work, LeNet-5 used gradient-based backpropagation and was capable of handling more complex and varied inputs, marking a major advance in document image recognition.

Classification of Characters

Can handle complex cases of handwritten digits

More complex CNN with 3 layers



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Foundational CNN (3/3)

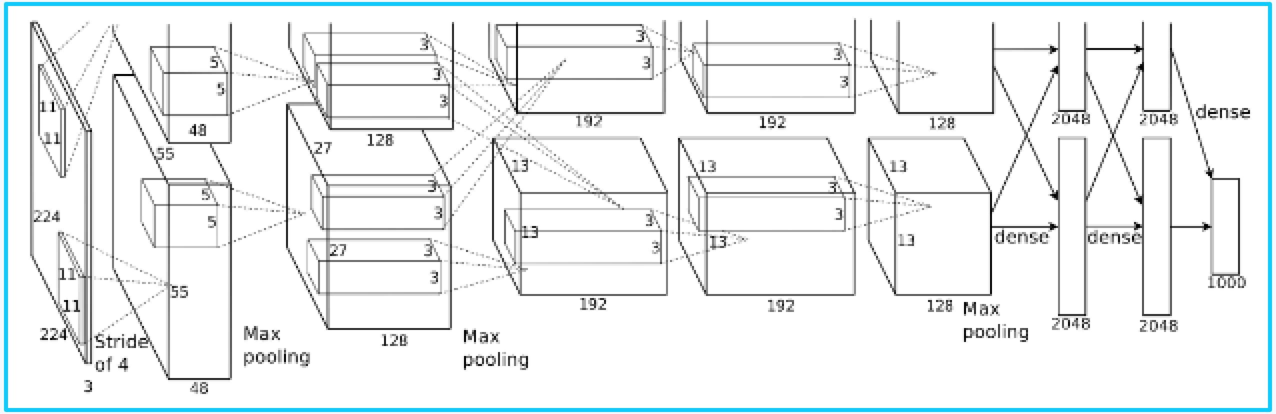
Paper - ImageNet Classification with Deep Convolutional Neural Networks

ImageNet Classification with Deep Convolutional Neural Networks, introduced by **Alex Krizhevsky, Ilya Sutskever, and Geoffrey Hinton** in 2012, revolutionized image classification by achieving record-breaking accuracy on the **ImageNet** competition. Known as **AlexNet**, the **8-layer CNN** showcased the power of deep learning, using ReLU activations, dropout, and GPU acceleration. It marked a turning point in computer vision, proving how effective CNNs could be for large-scale visual recognition tasks.

Image Classification with high complexity

Larger Network with 8-layers

First to use GPU for AI Training



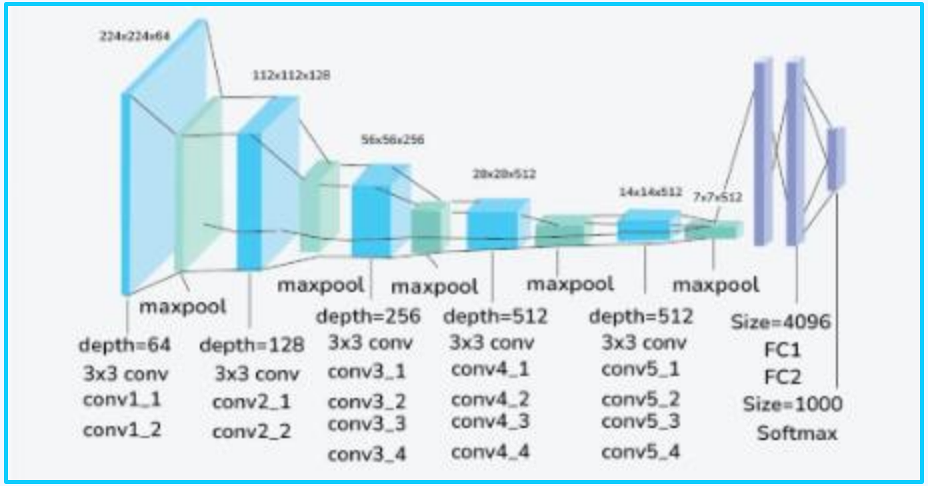
Deeper Network (1/2)

Paper - Very Deep CNN for Large-scale Image Recognition

VGGNet, introduced in 2014 by Simonyan and Zisserman, demonstrated that **deeper, simpler, and uniform CNN architectures** using only **3×3 convolutions** could significantly improve performance. With up to 19 layers, VGGNet set new benchmarks in image classification, influencing the design of many modern deep learning models.

Ready for Transfer Learning

Simplicity and Effectiveness



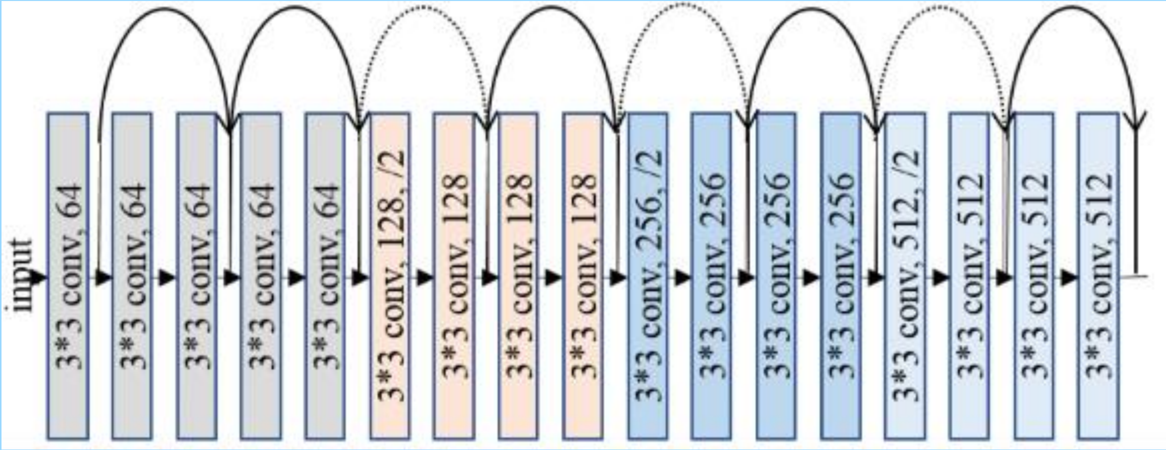
Deeper Network (2/2)

Paper - Deep Residual Learning for Image Recognition

Residual Networks (ResNets), developed by **Microsoft Research** in 2015, introduced shortcut connections that let layers learn residuals instead of full mappings. This made deep networks easier to train and more accurate. Models like **ResNet-50** stack these residual blocks to achieve great depth without performance loss.

Very Deep NN with more than 100 layers

Residuals enable deeper networks without accuracy loss



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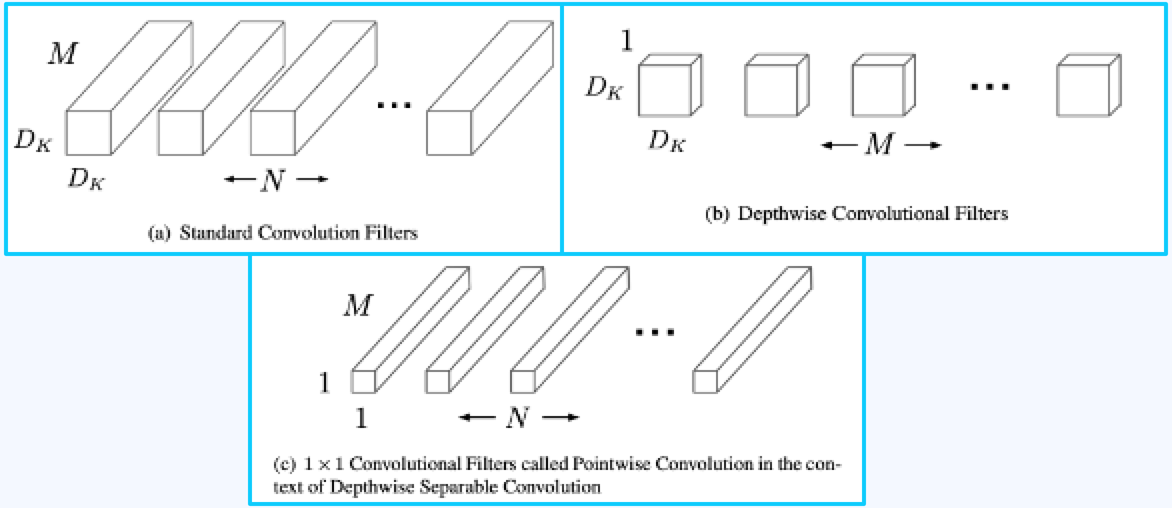
Efficiency-focused Model (1/2)

Paper - MobileNets: Efficient CNNs for Mobile Vision Applications

MobileNet, developed by Google, is a lightweight CNN designed for mobile and edge devices. Using **depthwise separable convolutions**, it achieves fast, efficient performance with minimal computational cost—ideal for real-time vision tasks on limited hardware.

Depthwise convolution & Pointwise convolution

Quantization & Weight Pruning



(a) Standard Convolution Filters

(b) Depthwise Convolutional Filters

(c) 1×1 Convolutional Filters called Pointwise Convolution in the context of Depthwise Separable Convolution

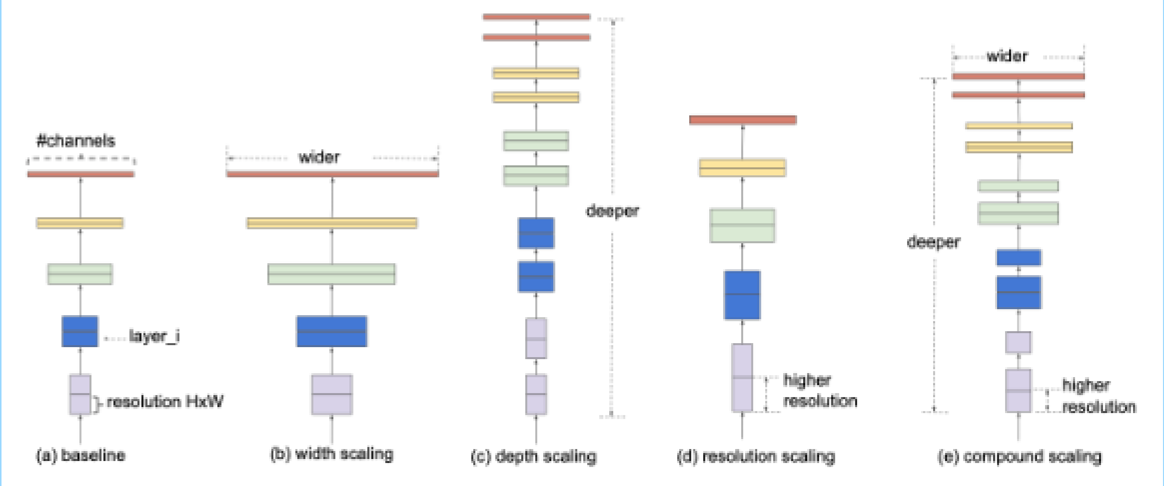
Efficiency-focused Model (2/2)

Paper - EfficientNet: Rethinking Model Scaling for CNNs

EfficientNet, introduced by Google in 2019, is a family of CNN models that balances accuracy and efficiency through **compound scaling**—jointly optimizing network depth, width, and resolution. It outperforms many larger models while using fewer parameters and less computation, making it ideal for both cloud and edge applications.

Compound Scaling

Built based on MobileNet



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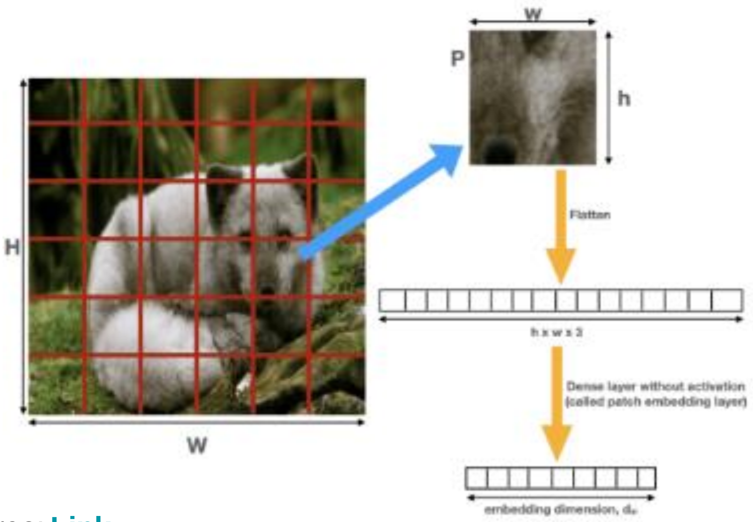
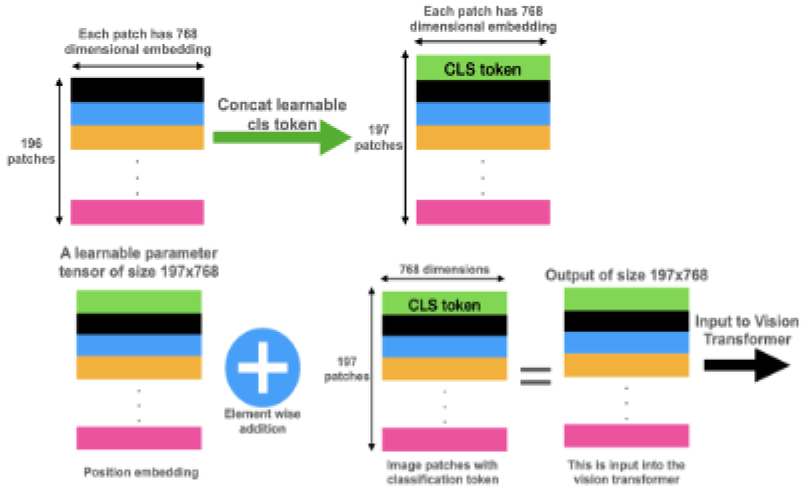
Vision Transformer (1/2)

Vision Transformer (ViT) is a deep learning model that applies the transformer architecture to image recognition tasks. It first splits an image into fixed-size **patches** (e.g., 16×16), flattens them, and embeds each as a token. A special **[CLS] token** is prepended to the sequence, and **positional embeddings** are added to retain spatial information. This sequence is then processed by a standard transformer, enabling the model to capture global context across the entire image.

1. Patches Generation



2. Packing Input Vectors



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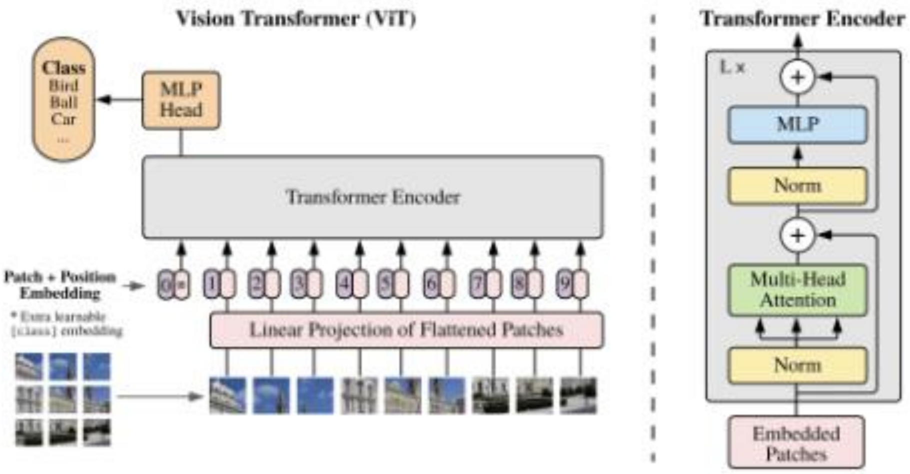
Vision Transformer (2/2)

3. Inference Through Transformer: The token sequence is passed through multiple **transformer encoder layers**, each consisting of:

- Layer normalization,
- Multi-head self-attention,
- Residual (skip) connections,
- Feed-forward neural network (MLP).

4. **Classification Head:** After encoding, the output corresponding to the **[CLS] token** is passed to a **classification head** (usually a fully connected layer) to generate the final prediction.



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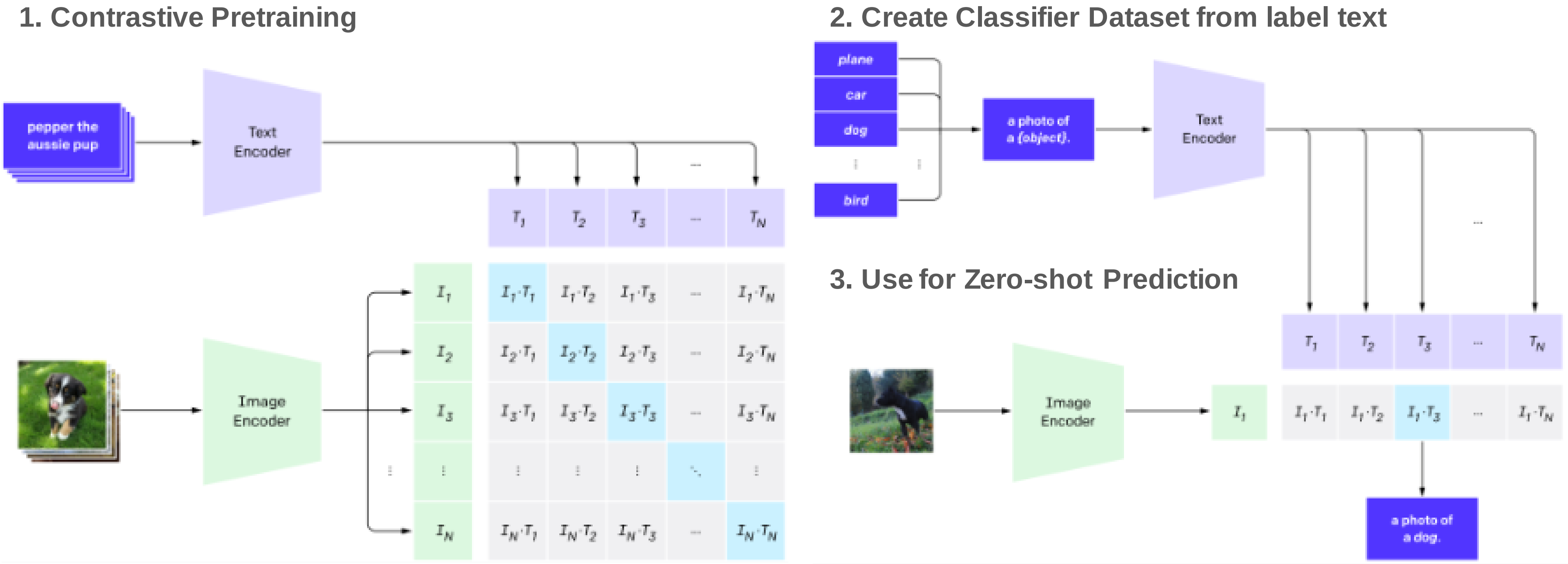
Contrastive Language-Image Pretraining

CLIP is a **multi-modal model** developed by **OpenAI** that learns to understand images and text **jointly**. It is trained to associate **images and their textual descriptions** without needing task-specific labels.



Bootstrapping Language-Image Pretraining

BLIP, developed by **Salesforce Research** in 2022, is a **vision-language model** designed to improve performance on tasks like **image captioning**, **visual question answering (VQA)**, and **image-text retrieval**. It uses both **vision and language encoders** in a unified architecture and focuses on making better use of web-scale noisy image-text data.

No.	Compared Items	CLIP	BLIP
1	**Main Goal**	Vision-language **alignment** (image–text matching)	Vision-language **understanding & generation**
2	**Architecture**	Dual encoder: image + text → independent embeddings	Dual encoder + fusion decoder for deep interaction
3	**Tasks Supported**	Zero-shot classification, image–text retrieval	Captioning, VQA, retrieval, multi-modal chat
4	**Text Output**	No generation	Can generate captions, answers
5	**Language Model Use**	No LLM; just transformers	Uses transformers (and in BLIP-2: LLMs like Flan-T5)

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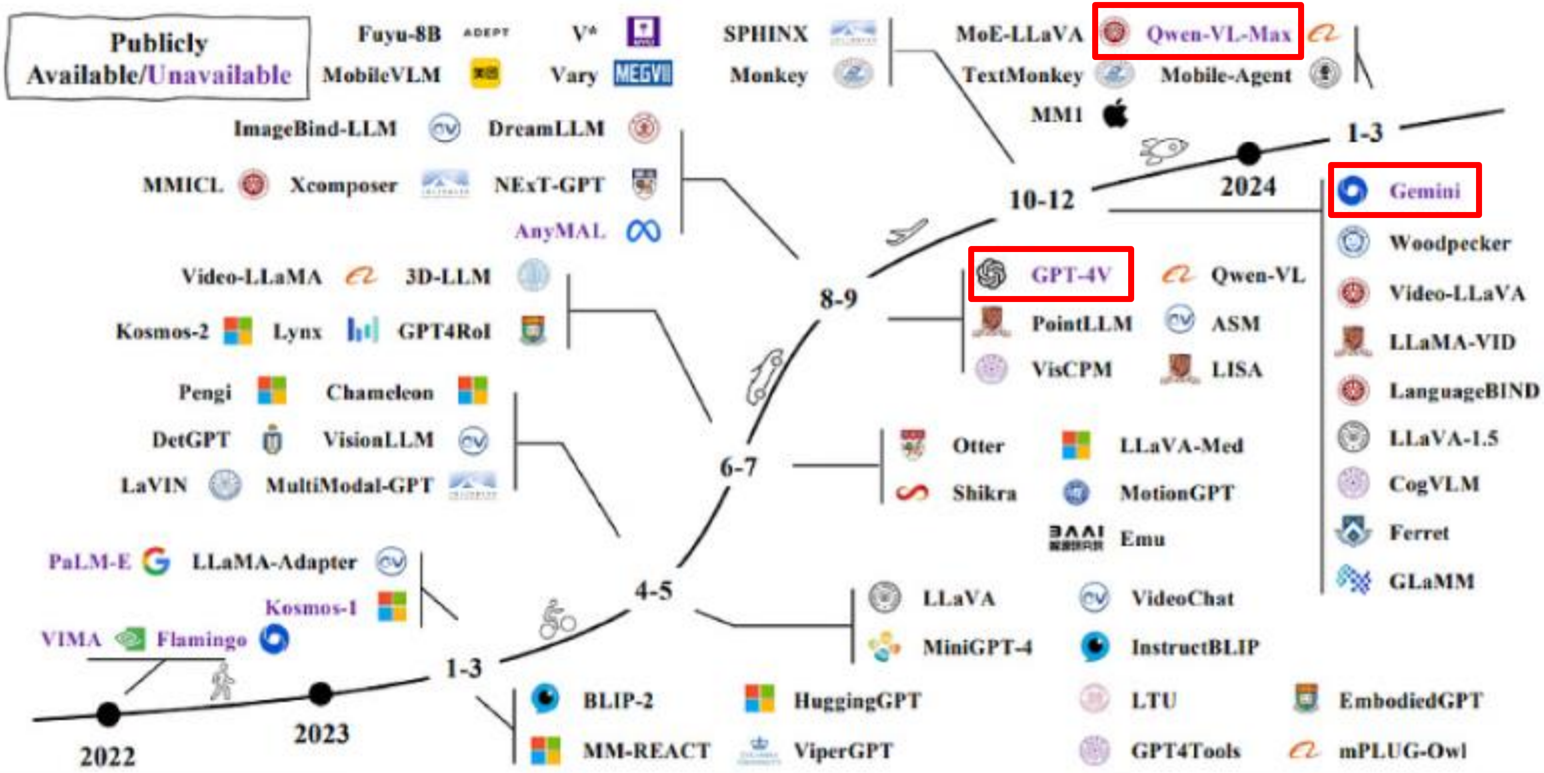
Multi-modal LLM



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Key Takeaways

Bigger model does not mean better

Each problem will have their own CV approach

Watch out for future of CV: Vision-Action Model

Applications

for General - Safety and Security

License plate detection

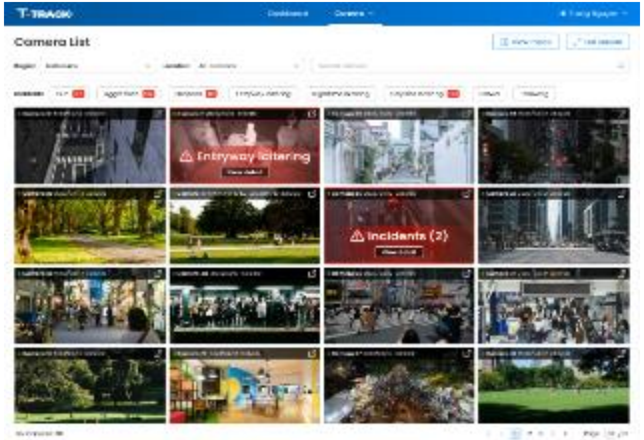
Loitering detection

Crowd detection

Nighttime unusual activity detection

Vandalism & throwing detection

Aggression detection



Smart City

Offices

Factories/ Industrial Parks

Commercial Buildings

Public Areas

AI Tech: Yolo, MediaPipe

for General - Employee Access Control

Identify QR codes/ faces, magnetic cards of employee

Open the door automatically

Mask check and automatic body temperature check

Enhanced security

Faster & contactless entry

Scalability

AI Tech: Yolo, FaceNet

for Agriculture - Pest Detection

Early identification and warnings enable optimized pesticide use

Supports both cloud and on-premise hosting

Apply Computer Vision for pest detection (up to 88% detection accuracy)

Provide precise identification of pests and diseases

Reduce farmer effort in farming work

Help farmers in selecting the correct pesticides

AI Tech: Yolo

for Agriculture - Pig Abnormal Activities Detection

Integrate real-time monitoring for pig counting and behavior tracking

Deploy edge processing technology

Detect early abnormal behaviors and disease through computer vision

Reduce mortality rates and improved animal health

Reduce labor costs and errors

Improve operational efficiency

AI Tech: Yolo, MediaPipe

for Manufacturing - Product Sorting/Counting

Automated product counting

Identify and categorize different product types based on shape, size, and other visual features

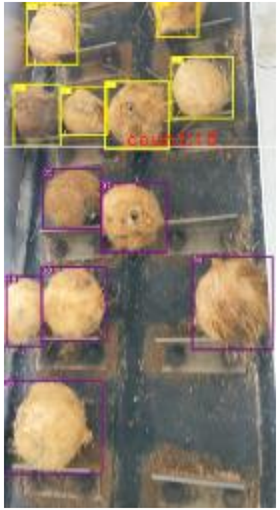
Integrate with robotic arms and conveyor systems

High accuracy & Speed

Reduce manual effort

Improve quality control

AI Tech: Yolo



for Manufacturing - Machinery Data Extraction

Extract and digitize information from industrial control screens

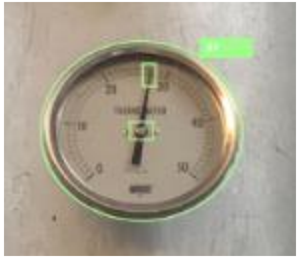
Real-time monitoring

Generate performance reports and trend analysis

High accuracy & Speed

Reduce manual effort

AI Tech: Yolo, PaddleOCR



for Warehouse, Manufacturing - PPE Detection

Real-Time PPE violation alerts

Edge AI for on-site processing

Smart reporting & insights

Enhance workplace safety

Reduces manual monitoring

Improve efficiency & productivity

AI Tech: Yolo


In factory


In warehouse

SC Sharing - Teledeaf

THANK YOU!